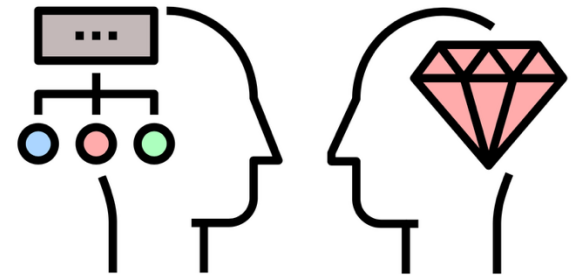


Machine Learning for Materials

8. Accelerated Discovery

Aron Walsh

Department of Materials
Centre for Processable Electronics

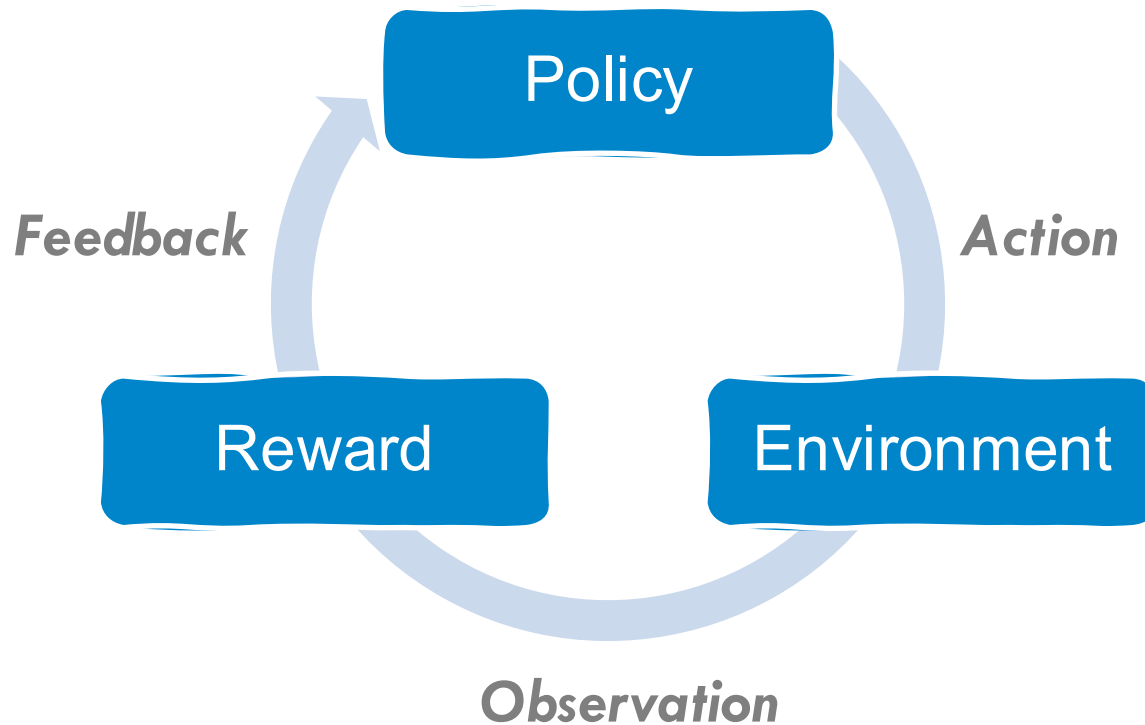


Module Contents

1. Introduction
 2. Machine Learning Basics
 3. Materials Data
 4. Crystal Representations
 5. Classical Learning
 6. Deep Learning
 7. Building a Model from Scratch
 - 8. Accelerated Discovery**
 9. Generative Artificial Intelligence
 10. Future Directions
-

Key Concept #8

Use data to choose the next best experiment (or simulation)



Example: “Choose next anneal temperature to maximise conductivity”

Every step balances system exploration and exploitation

“A *problem* in artificial intelligence is one which is so complex that it cannot be solved using any normal algorithm”

Class Outline

Accelerated Discovery

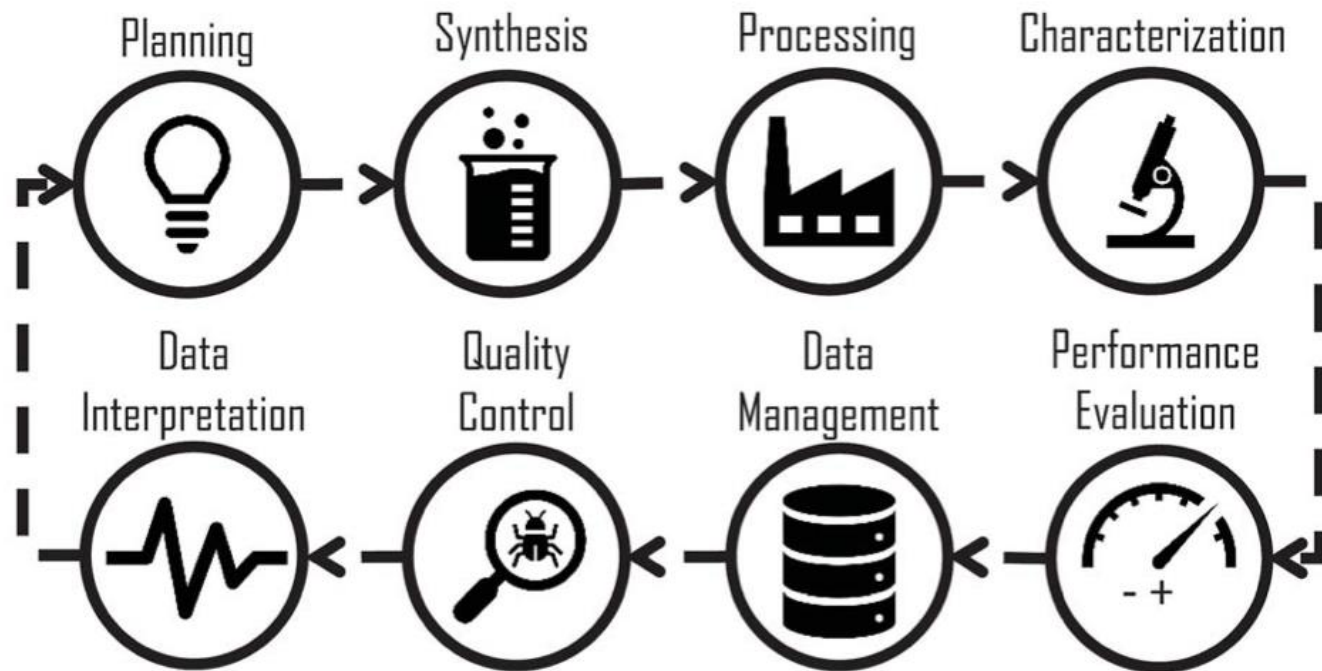
A. Automated Experiments

B. Bayesian Optimisation

C. Reinforcement Learning

Accelerate Scientific Discovery

Research can be broken down into a set of tasks that can each benefit from acceleration



Traditional
research
workflow

Accelerate Scientific Discovery

Research can be broken down into a set of tasks that can each benefit from acceleration



Automation



Parallelization



ML
Models



Data
Repositories



Active
Learning



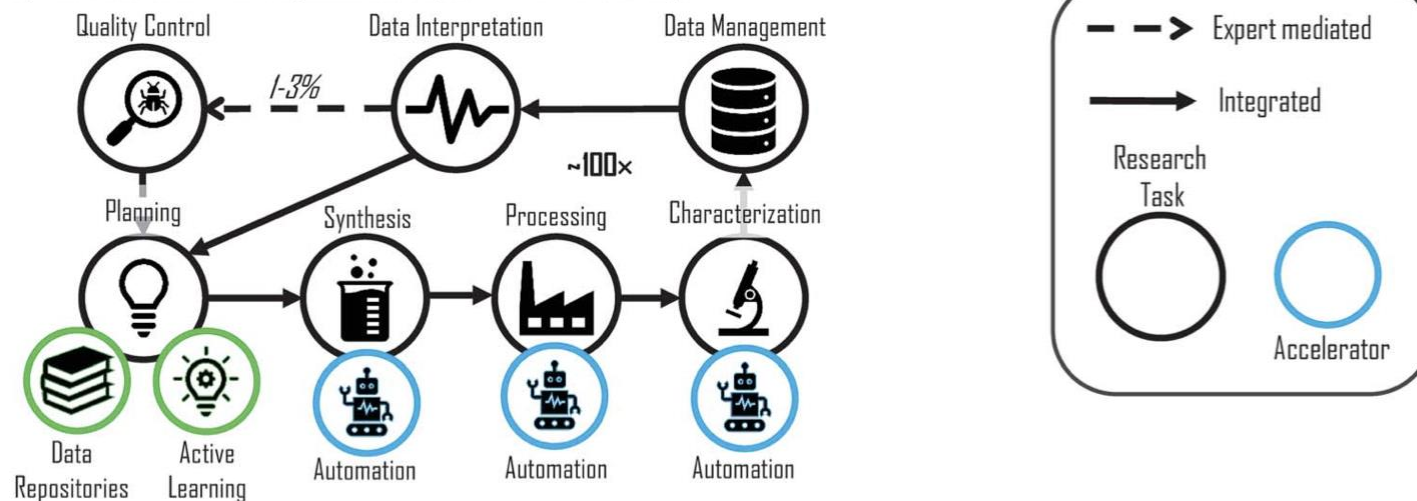
Automated
Reasoning

Potential
for
speedup

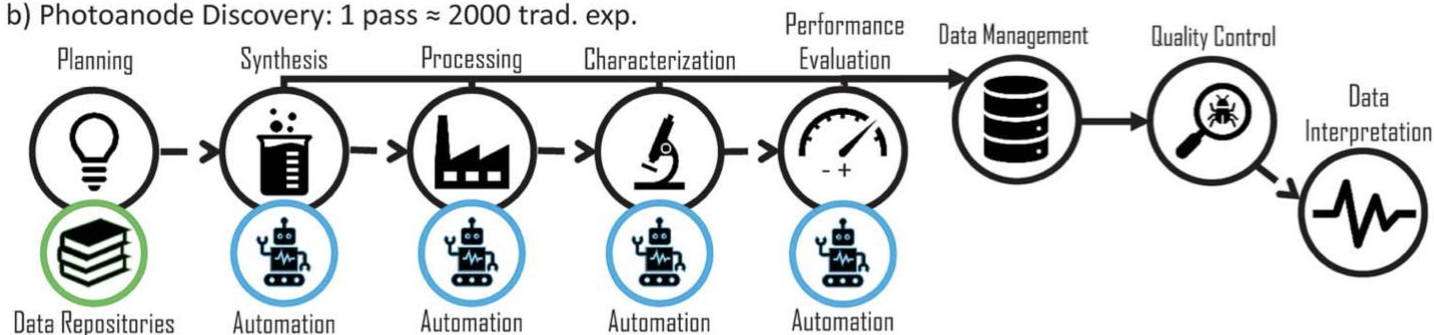
Accelerate Scientific Discovery

Workflow classification of published studies

a) Carbon Nanotube Synthesis: 1 pass $\approx$ 100 trad. exp.

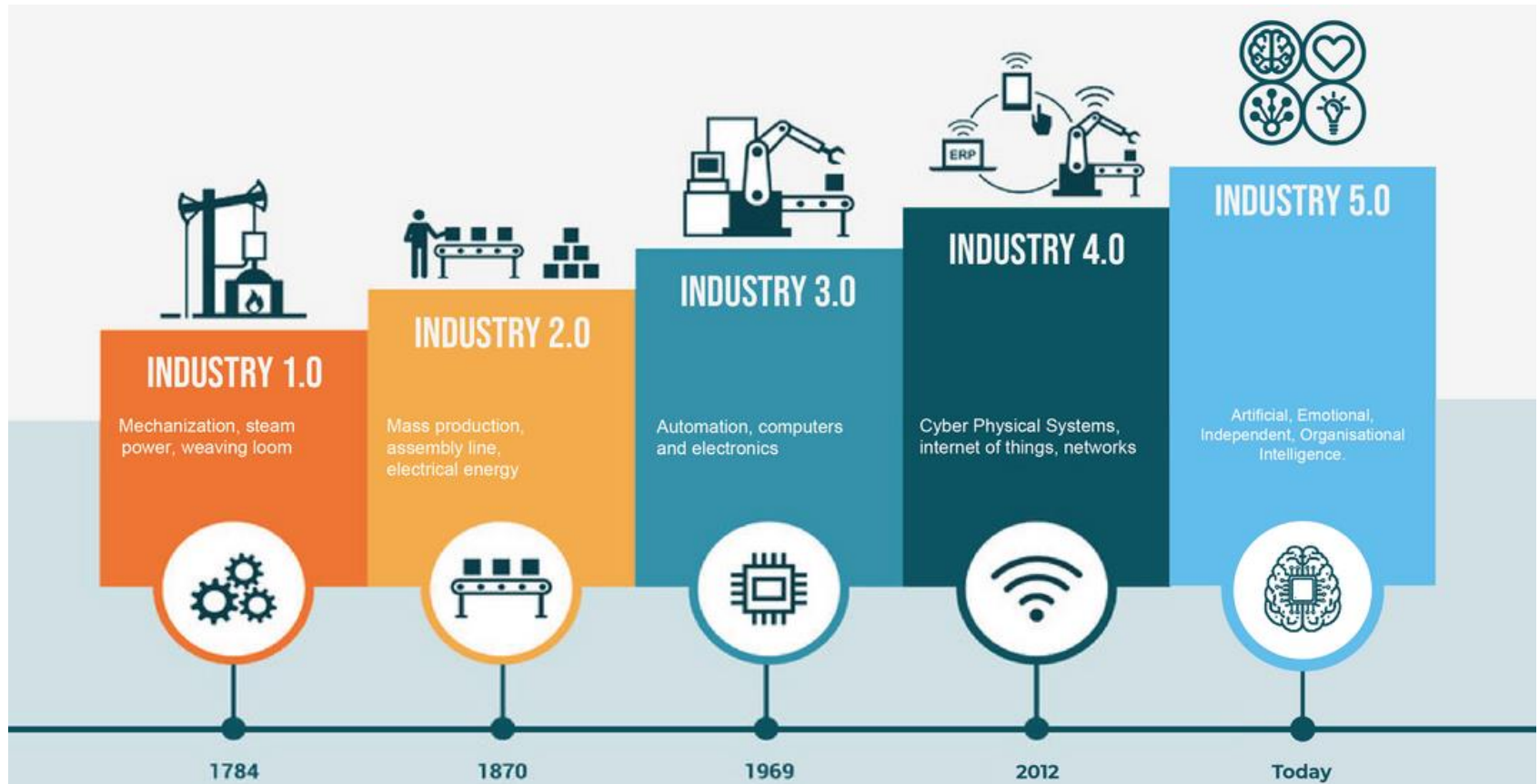


b) Photoanode Discovery: 1 pass $\approx$ 2000 trad. exp.



Automation and Robotics

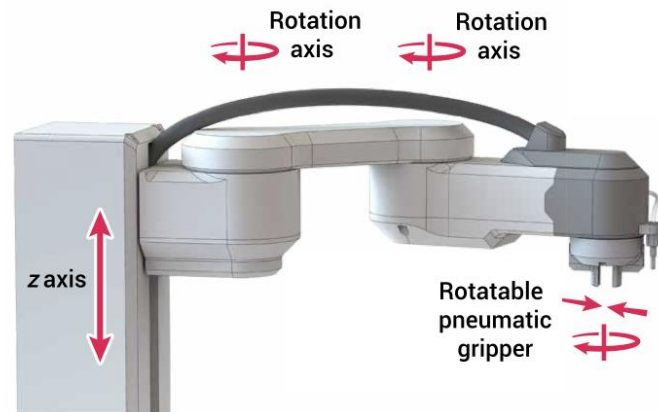
Execution of physical tasks to achieve a target using autonomous or collaborative robots



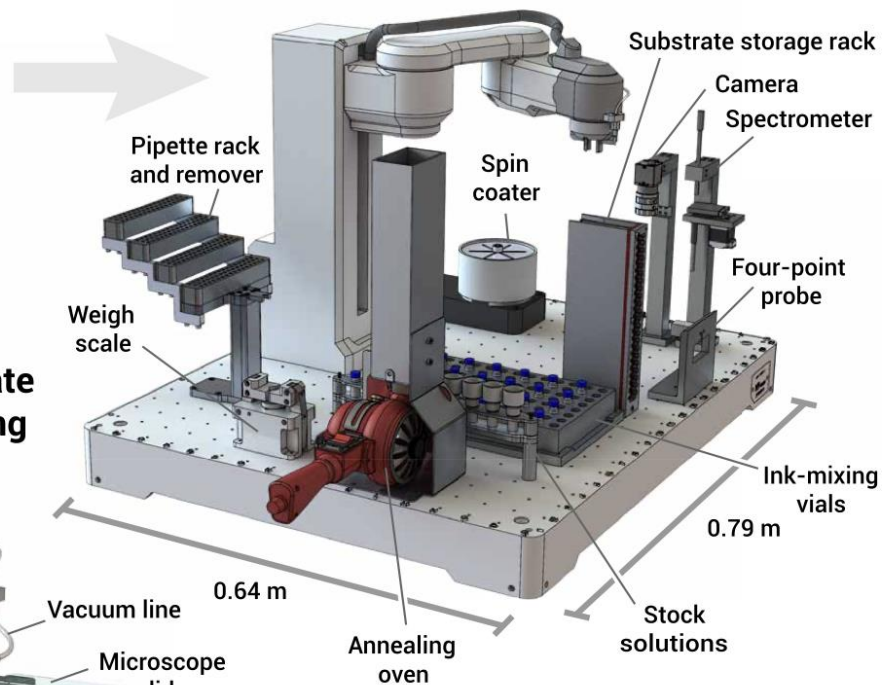
Automation and Robotics

Robots can be tailored for a wide range of materials synthesis and characterisation tasks

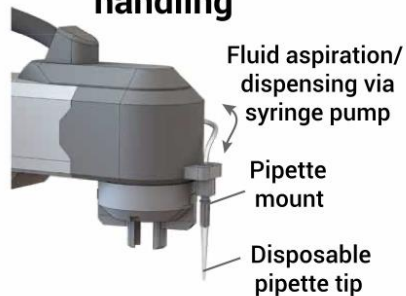
A Multipurpose robotic platform



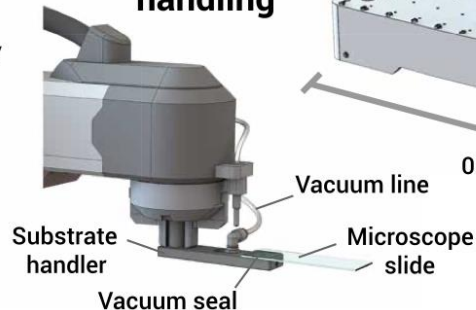
D Robotic platform configured for thin-film materials research



B Fluid handling



C Substrate handling



Automation and Robotics

Self-driving labs (SDL) are now operating



Flexible Automation Systems

Modular hardware with computer-controlled
synthesis and characterisation



Small Molecules



Materials

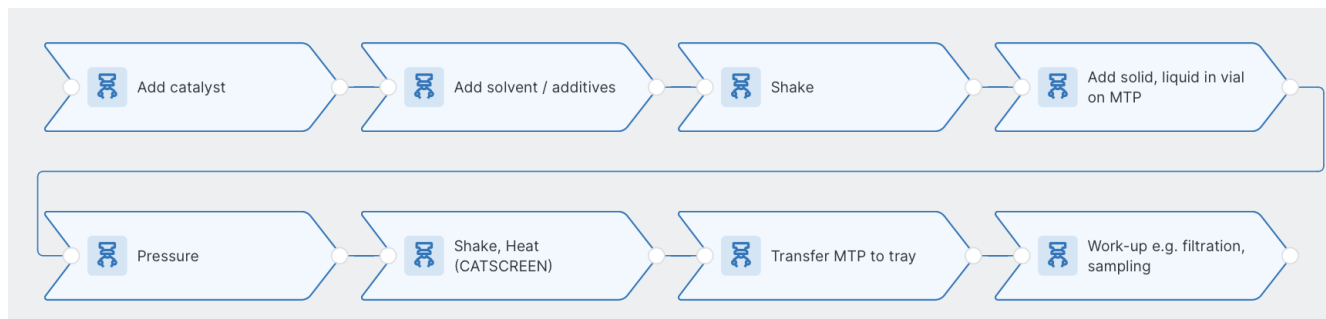


Applications

Flexible Automation Systems

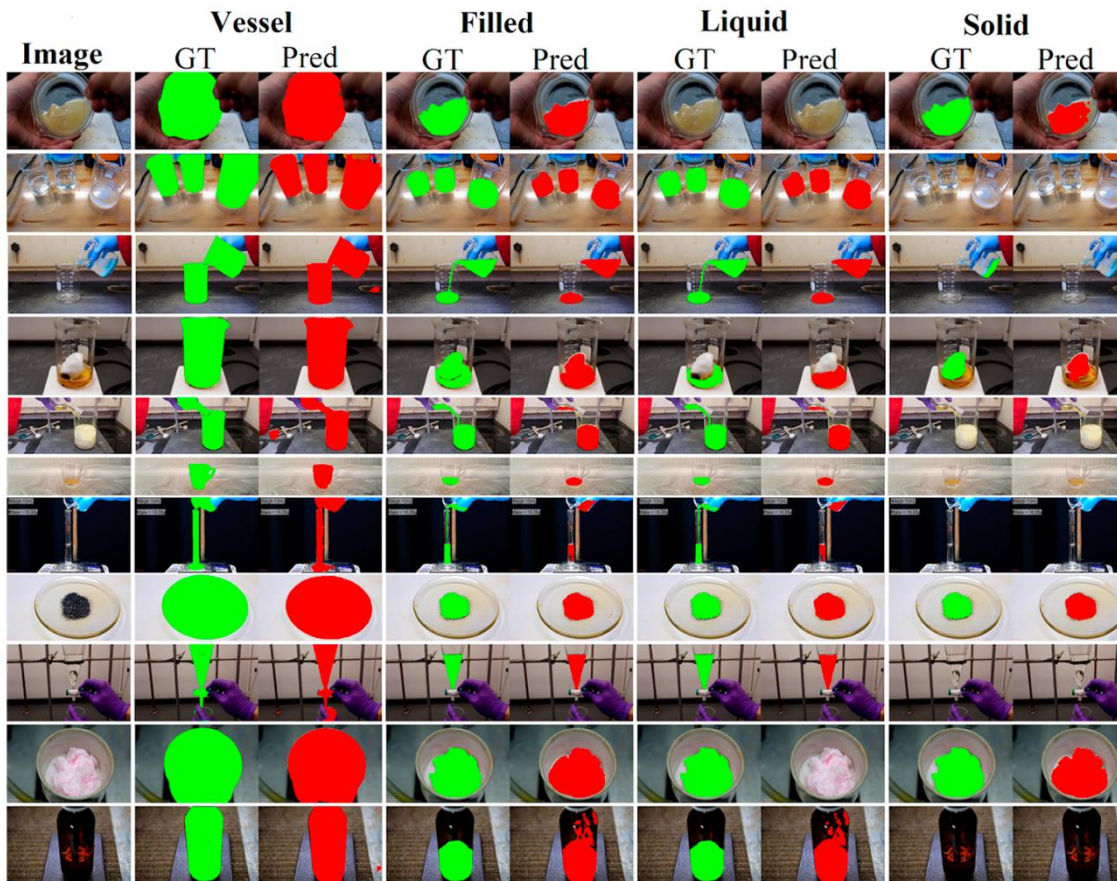
Automation platforms designed to deliver complex research workflows (fixed platform or mobile)

Usually a mix of proprietary code, with GUI and Python API for user control



Automation and Robotics

Robots can be equipped with sensors and artificial intelligence to interact with their environment



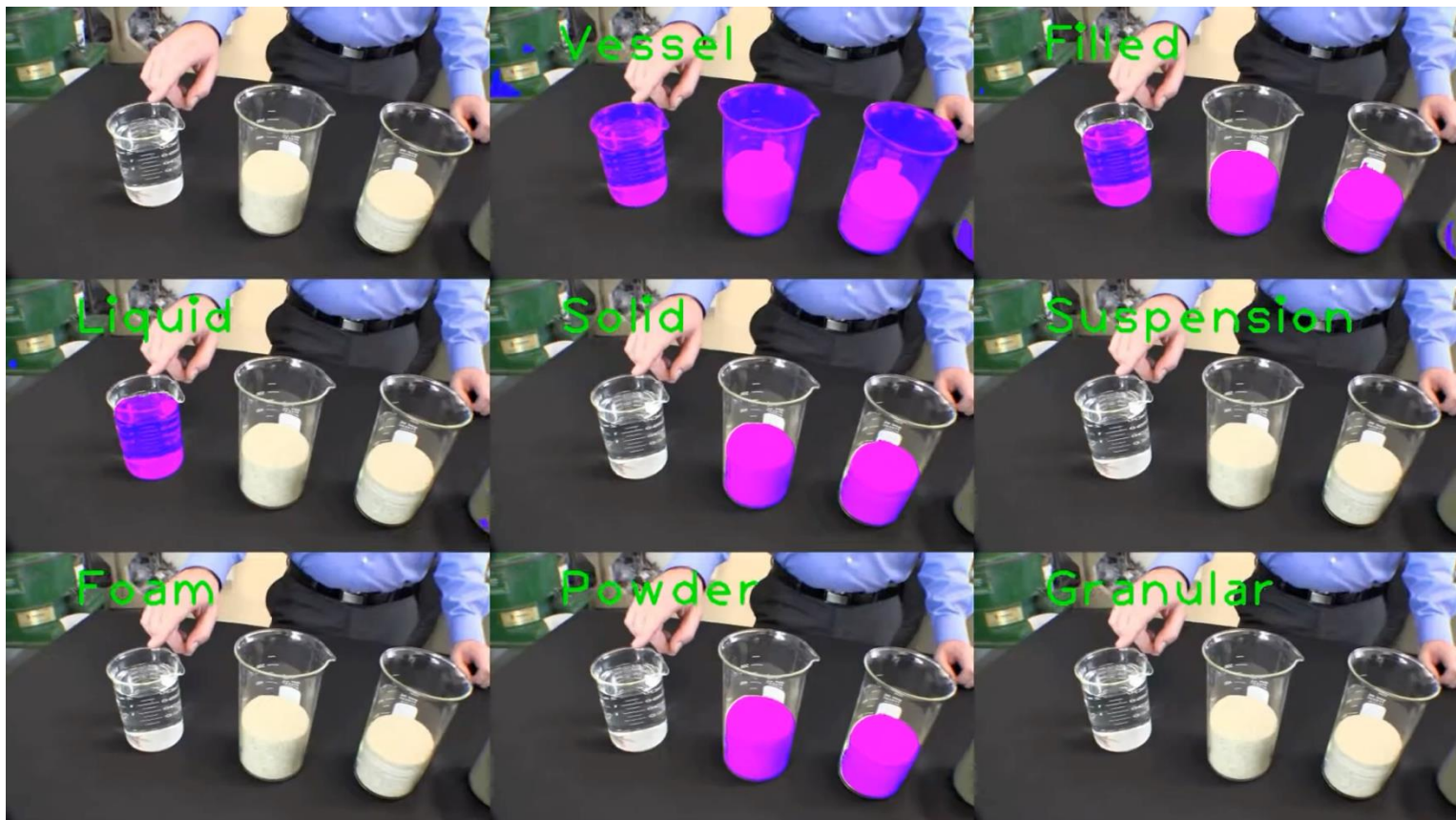
Adapting computer vision models for laboratory settings

GT = ground truth

Pred = predicted

Automation and Robotics

Robots can be equipped with sensors and artificial intelligence to interact with their environment



Optimisation

Algorithms to efficiently achieve a desired research objective. Considerations:

Objective function (O): Materials properties or device performance criteria, e.g. battery lifetime

Parameter selection: Variables that can be controlled, e.g. temperature, pressure, composition

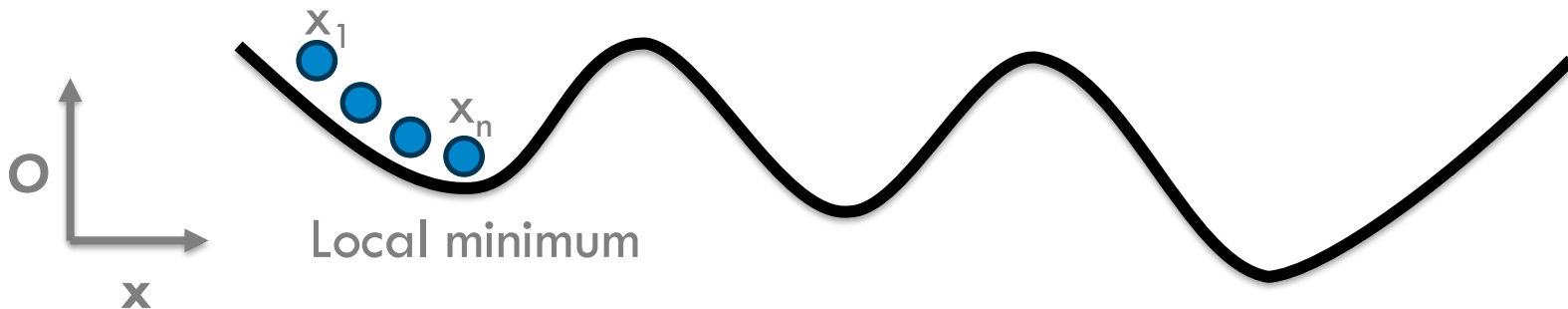
Data acquisition: How the data is collected, e.g. instruments, measurements, automation

Optimisation Algorithms

Local optimisation – find the best solution in a limited region of the parameter space ($\mathbf{x}$)

Gradient based: iterate in the direction of the steepest gradient (∇O), e.g. gradient descent

Hessian based: use information from the second derivatives ($\nabla^2 O$), e.g. quasi-Newton



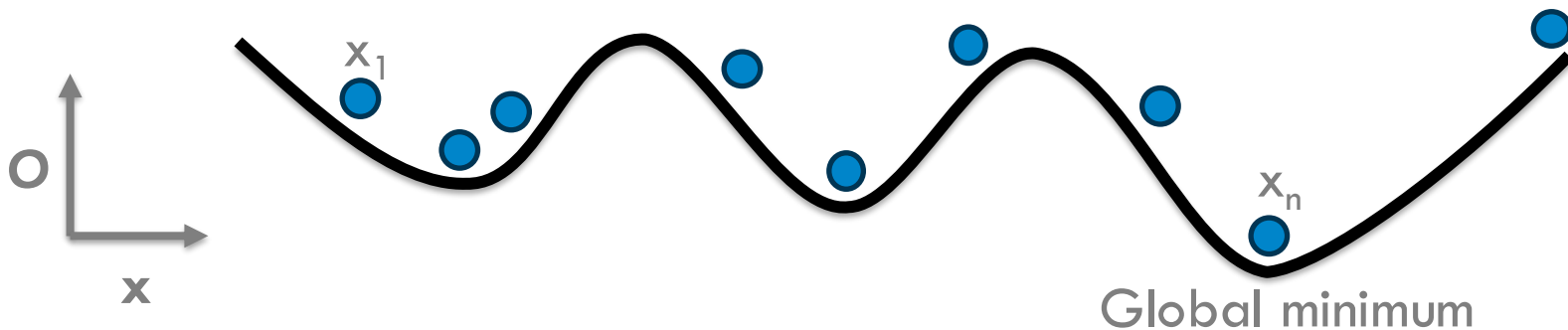
The same concepts are involved in ML model training

Optimisation Algorithms

Global optimisation – find the best solution from across the entire parameter space

Numerical: iterative techniques to explore parameter space, e.g. downhill simplex, simulated annealing

Probabilistic: incorporate probability distributions, e.g. Markov chain Monte Carlo, Bayesian optimisation



The same concepts are involved in ML model training

Class Outline

Accelerated Discovery

A. Automated Experiments

B. Bayesian Optimisation

C. Reinforcement Learning

Bayesian Optimisation (BO)

Use prior (measured or simulated) data to decide which experiment to perform next
(parameters to sample)

Probabilistic (Surrogate) Model

Approximation of the true objective function

$O(\mathbf{x}) \sim f(\mathbf{x})$, e.g. Gaussian process, $GP(\underset{\text{new}}{\mathbf{x}}, \underset{\text{known}}{\mathbf{x}'})$

Acquisition Function

Selection of the next sample point, e.g.
upper confidence bound (UCB), probability of
improvement (PI), expected improvement (EI)

Bayesian Optimisation (BO)

Use prior (measured or simulated) data to decide which experiment to perform next
(parameters to sample)

Probabilistic (Surrogate) Model

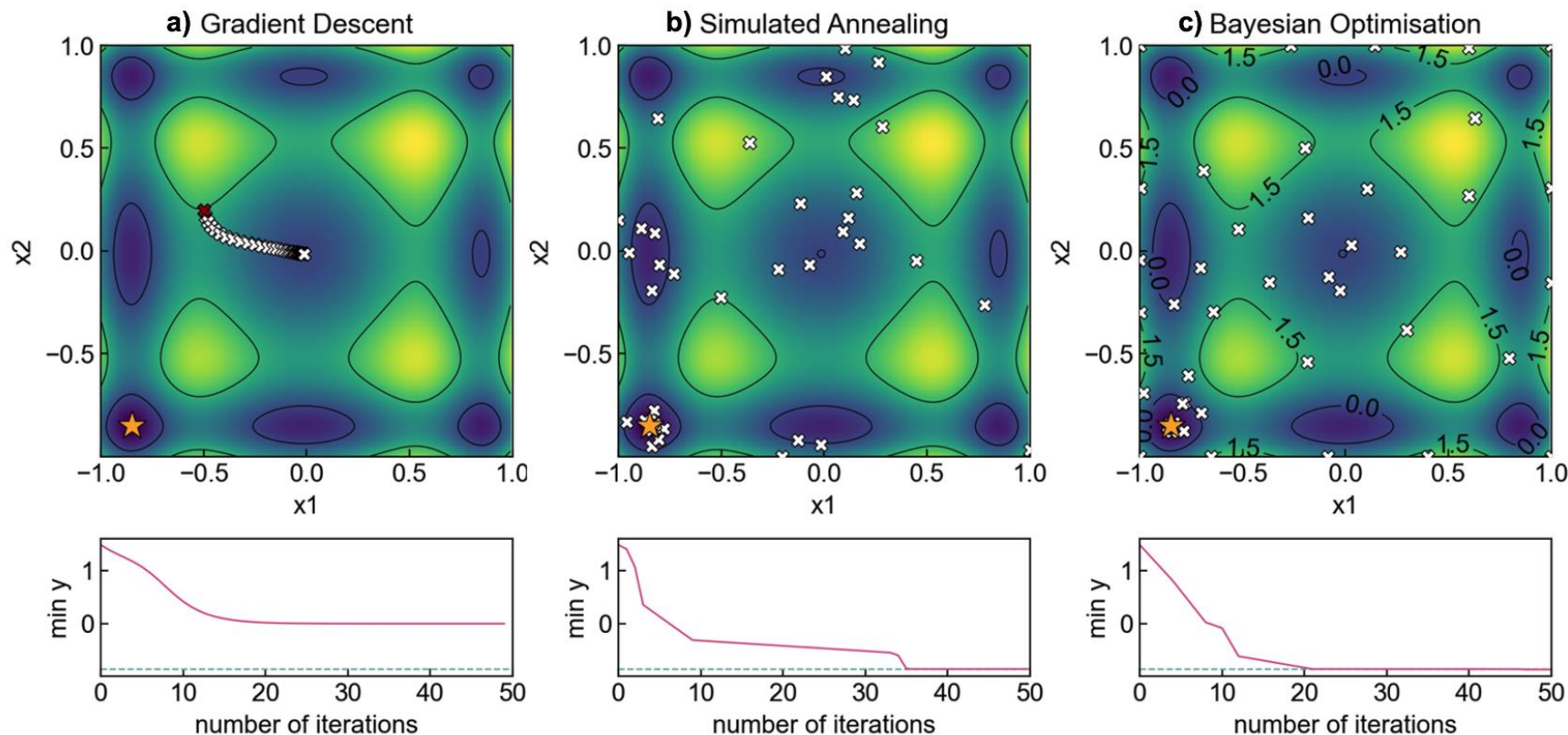
Gaussian process: $f(\mathbf{x}) \sim \text{GP}(\underbrace{\mu(\mathbf{x})}_{\text{mean function}}, \underbrace{k(\mathbf{x}, \mathbf{x}')}_{\text{Gaussian kernel function}})$

$k(\mathbf{x}, \mathbf{x}')$ measures the similarity points $\mathbf{x}$ and $\mathbf{x}'$
new known

- Kernel controls function smoothness and uncertainty
 - Similar $\mathbf{x}$ share information
- Dissimilar $\mathbf{x}$ default to the mean with high uncertainty

Bayesian Optimisation (BO)

Use prior (measured or simulated) data to decide which experiment to perform next



Exploration–Exploitation Tradeoff

Upper confidence bound (UCB) trades off
mean prediction and uncertainty

$$x_{\text{next}} = \max_x (\mu(x) + \beta \sigma(x))$$

What to
do next

Prediction based
on prior knowledge

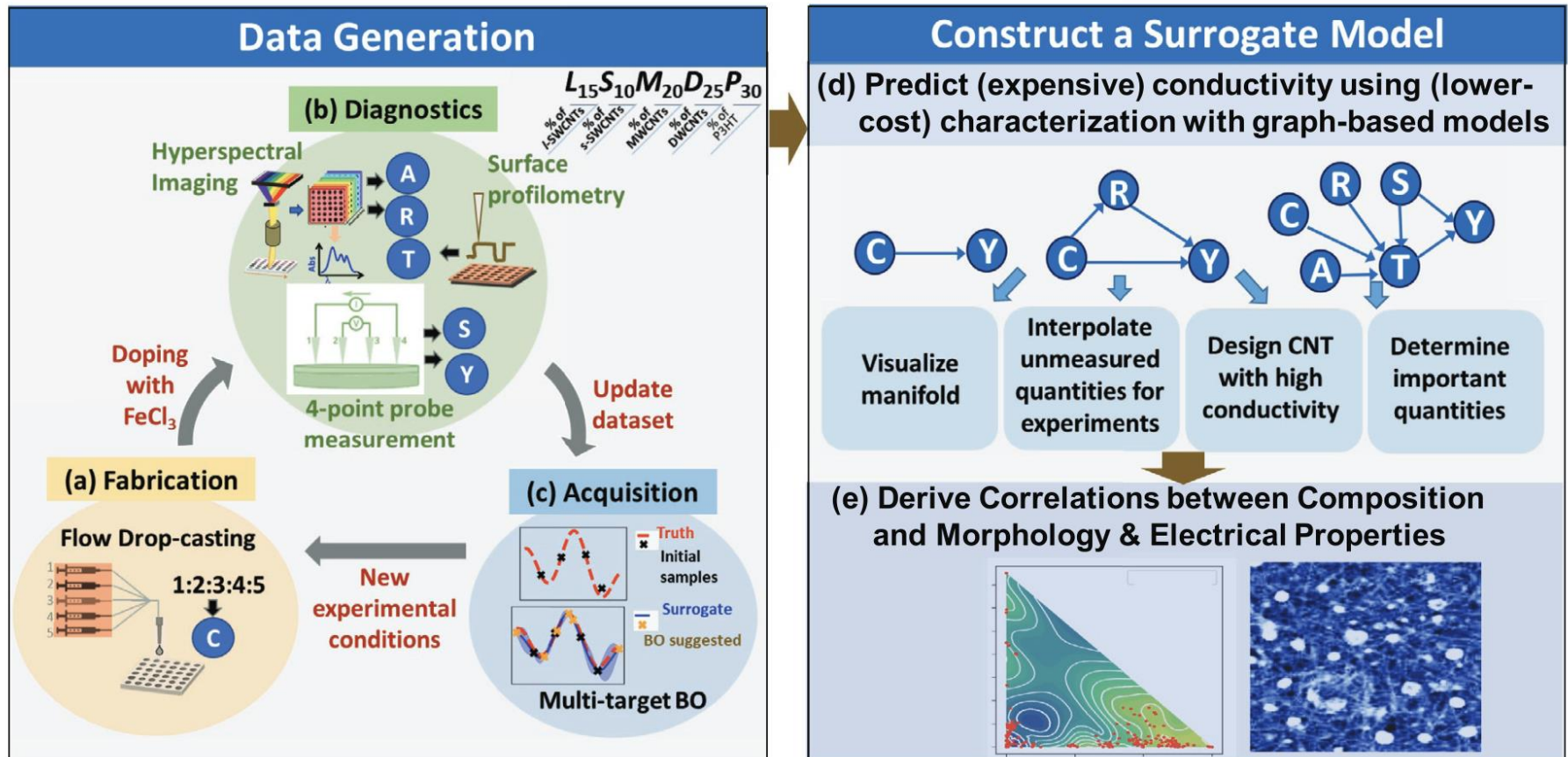
Weighted
Uncertainty

A tunable
(learnable)
hyperparameter
of UCB

$\beta < 1$ focus on exploitation
 $\beta \sim 1$ balance risk and reward
 $\beta > 1$ focus on exploration

Applications of BO

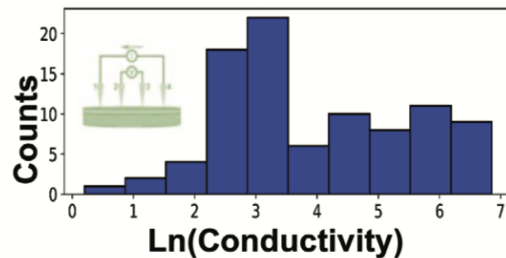
Maximise electrical conductivity of a composite (P3HT-CNT) thin-film



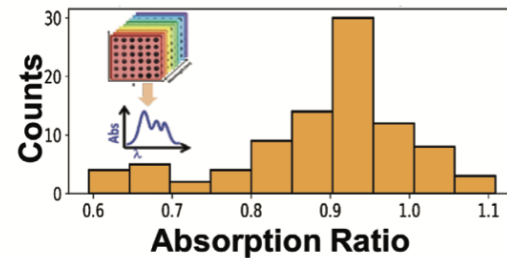
Applications of BO

Maximise electrical conductivity of a composite (P3HT-CNT) thin-film

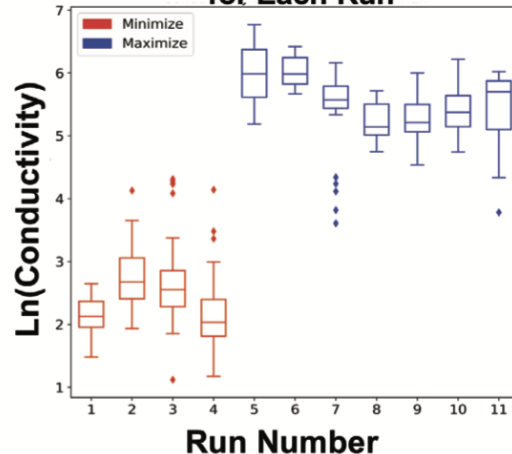
Conductivity Distribution from Initial Binary Combinations



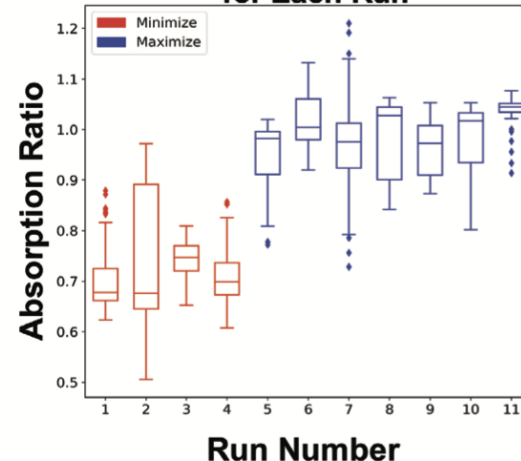
Absorption Ratio Distribution from Initial Binary Combinations



Conductivity Distribution for Each Run



Absorption Ratio Distribution for Each Run

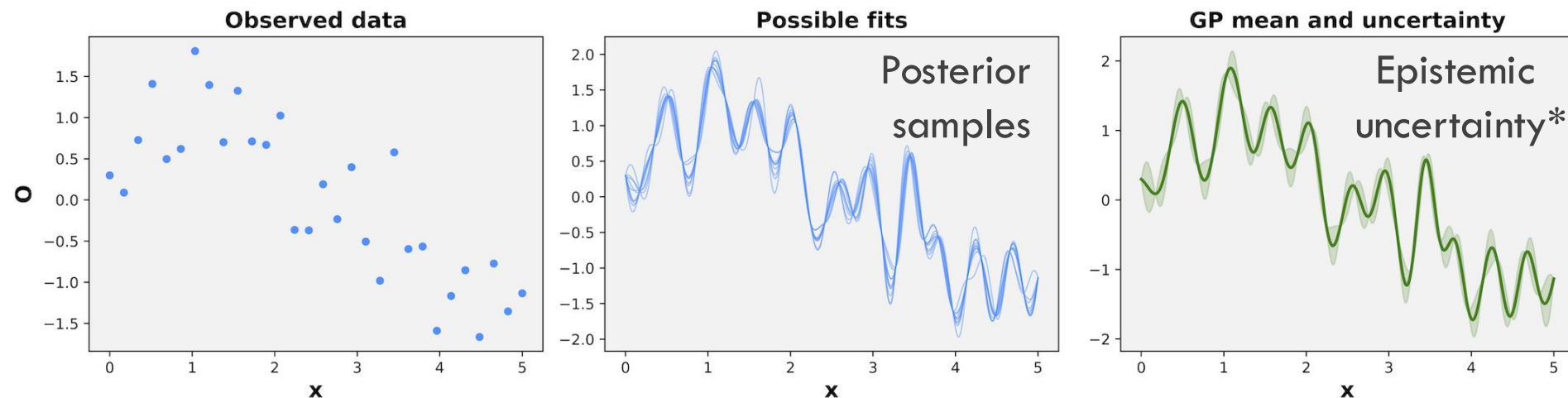


Active Learning (AL)

BO: find inputs that maximise the objective function

AL: find inputs that minimise uncertainty

Target unknown regions that can improve the model

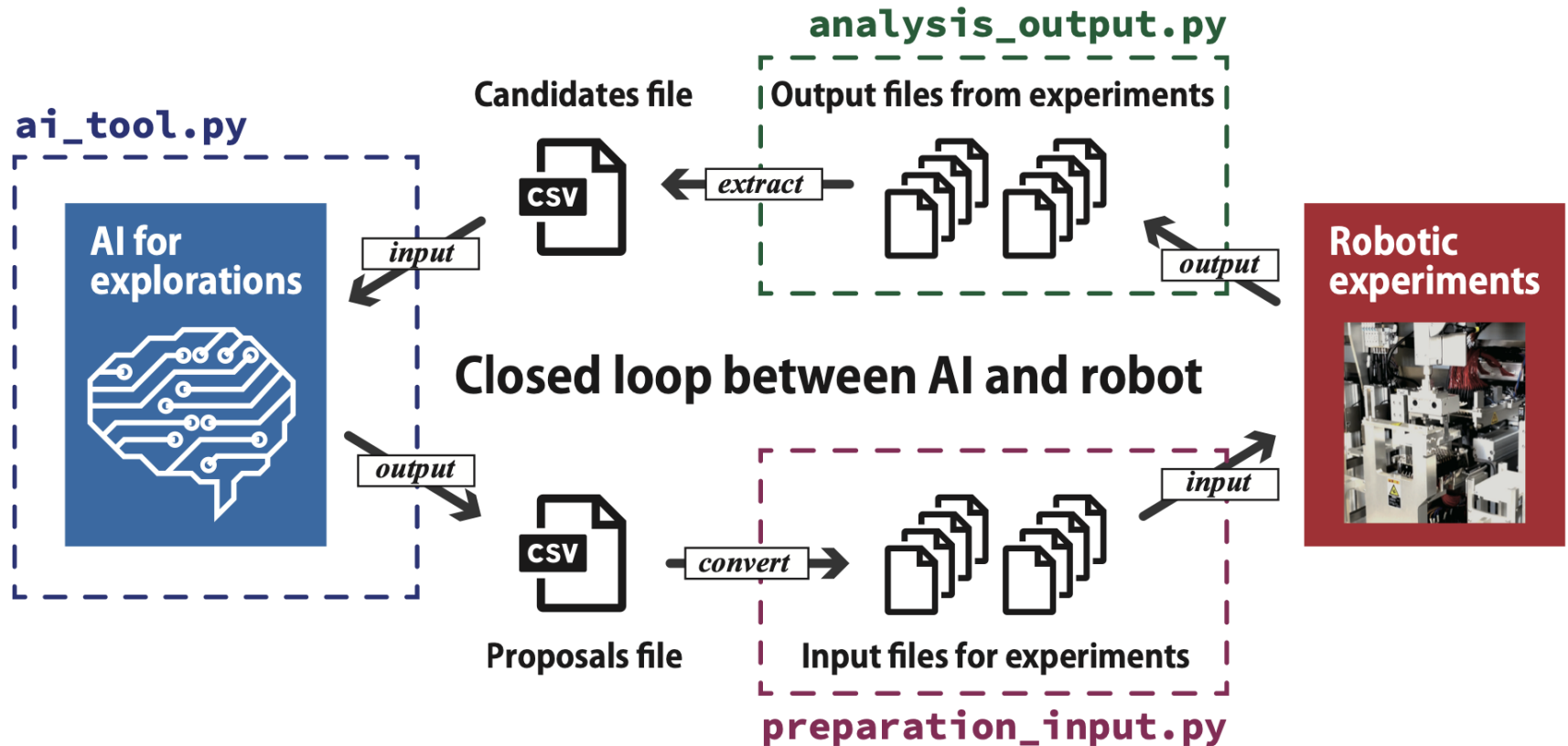


Gaussian process is updated with new observations to yield revised function values and uncertainty

*Epistemic = data-limited (reducible) uncertainty; Aleatoric = noise (irreducible)

Integrated Research Workflows

Feedback loop between optimisation model
and automated experiments



Integrated Research Workflows

Feedback loop between optimisation model
and automated experiments

Initial stage of the candidates file

descriptor 1	...	descriptor d	objective 1	...	objective l
1	...	0			
1	...	0.5			
1	...	1			
0.5	...	0			
0.5	...	0.5			
0.5	...	1			
0	...	0			
0	...	0.5			
0	...	1			

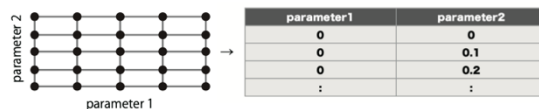
After three
experiments

Candidates file after some experiments

descriptor 1	...	descriptor d	objective 1	...	objective l
1	...	0			
1	...	0.5	12	...	20
1	...	1			
0.5	...	0			
0.5	...	0.5			
0.5	...	1	5	...	8
0	...	0			
0	...	0.5			
0	...	1	23	...	2

Examples for the list of descriptors

Discretized parameter space



Combination of materials

	A	B	C	D	E	F	...
ABC	1	1	1	0	0	0	...
BDF	0	1	0	1	0	1	...
CEF	0	0	1	0	1	1	...
:	:	:	:	:	:	:	...

Materials descriptors

	Number	MendeleevN umber	Atomic Weight	MeltingT	Column	Row	...
Fe _{0.3} Ni _{0.2} Mn _{0.2}	26.28	55.85	56.39	1703.85	8.28	4.0	...
Ag _{0.8} Cd _{0.2}	47.2	66.0	108.77	1106.78	11.20	5.0	...
Hf _{0.5} Ta _{0.5}	72.5	46.5	179.71	2898.0	4.5	6.0	...
:	:	:	:	:	:	:	...

e.g. by magpie

Fingerprint of molecules

	16	50	80	389	1088	1313	...
0	0	0	0	1	1	0	...
1	0	1	1	1	0	1	...
0	0	1	1	0	0	0	...
:	:	:	:	:	:	:	...

e.g. by Morgan fingerprint

Class Outline

Accelerated Discovery

A. Automated Experiments

B. Bayesian Optimisation

C. Reinforcement Learning

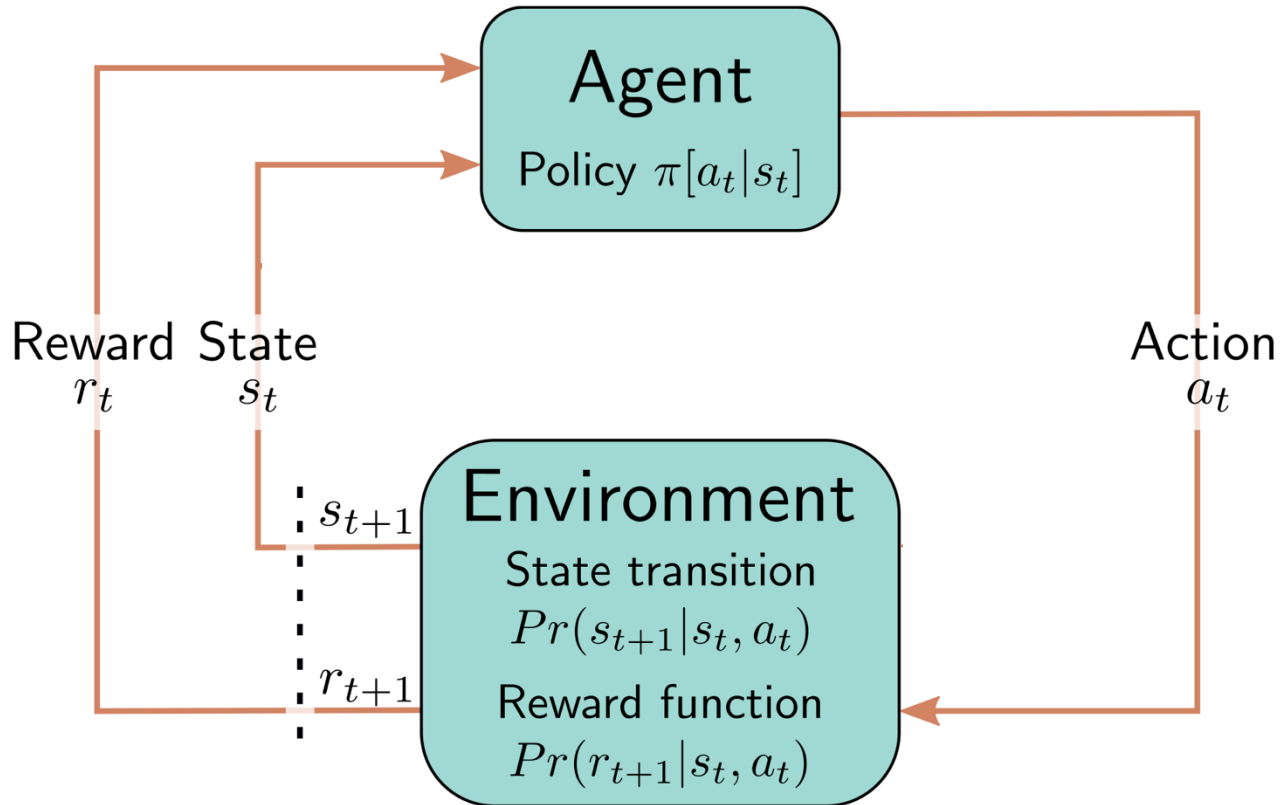
Reinforcement Learning (RL)

An agent interacts with an environment to learn decision-making strategies that achieve a specific goal

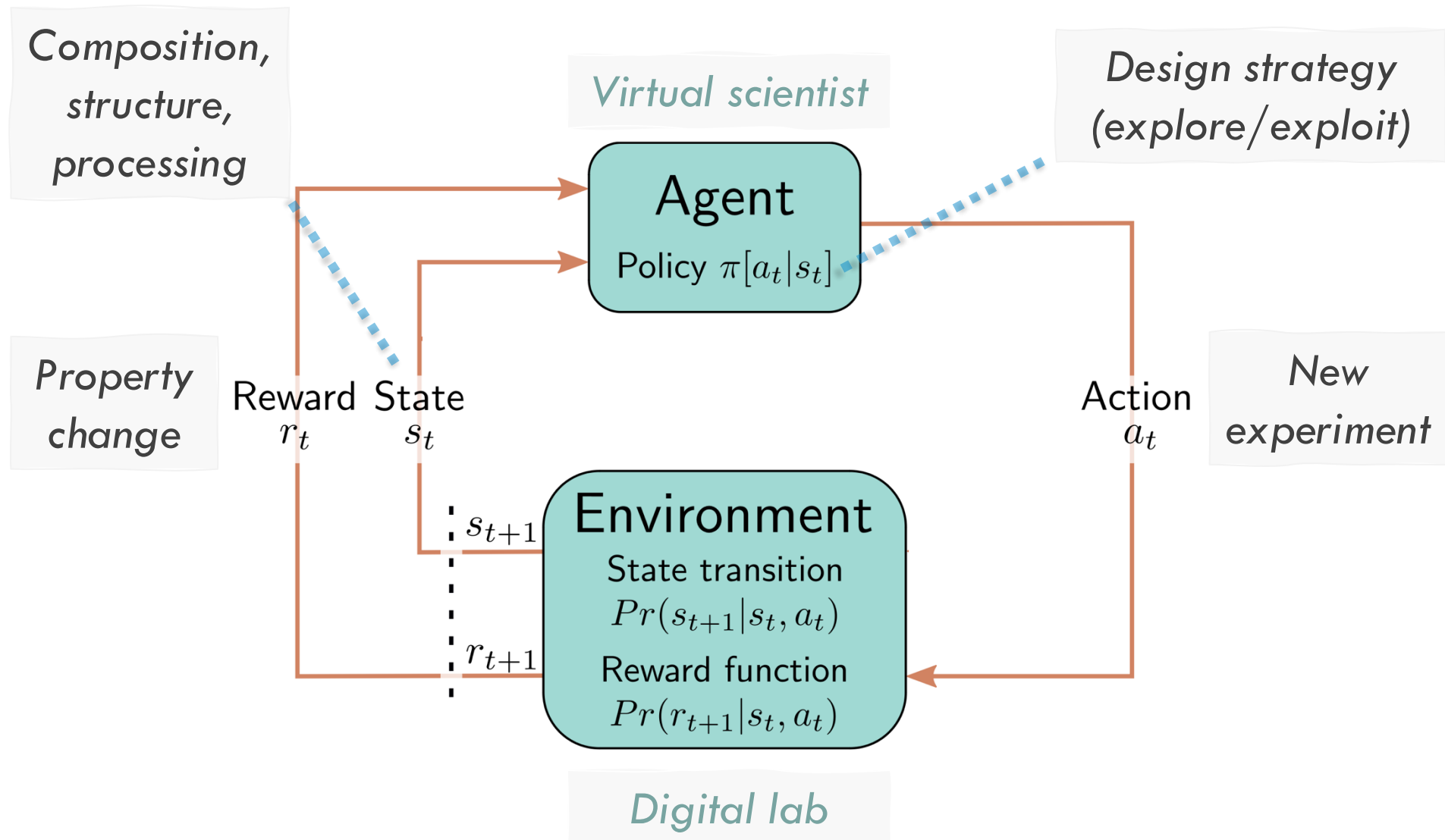


Early applications in video games (maximise score), finance (profit), and robotics (task completion)

Reinforcement Learning (RL)



Reinforcement Learning (RL)



RL Policy

Data-driven decision making that adapts over time

Probability of action a_t given state s_t

Expected reward for action a_t

$$\pi(a_t | s_t) = \frac{\exp(Q(s_t, a_t) / \tau)}{\sum_{a' \in \mathcal{A}} \exp(Q(s_t, a') / \tau)}$$

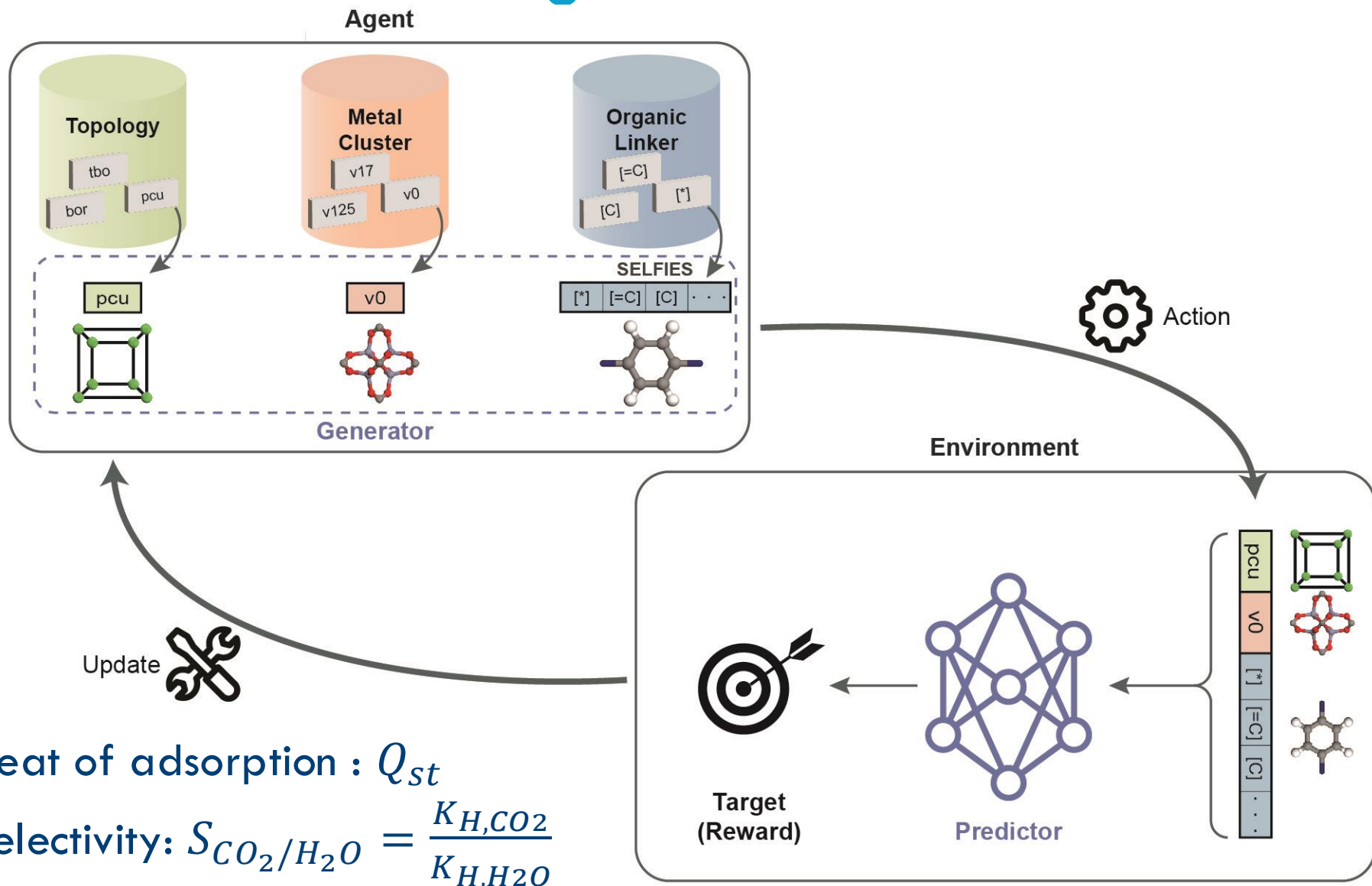
Sum over all possible actions

Effective temperature for exploration/exploitation balance

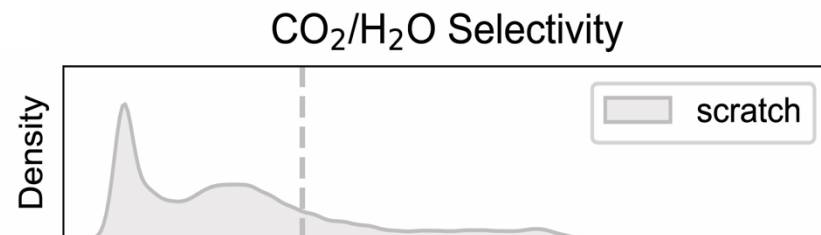
This familiar equation is a softmax (Boltzmann) policy

$Q(s, a)$ is the expected reward from taking action a in state s , following policy π

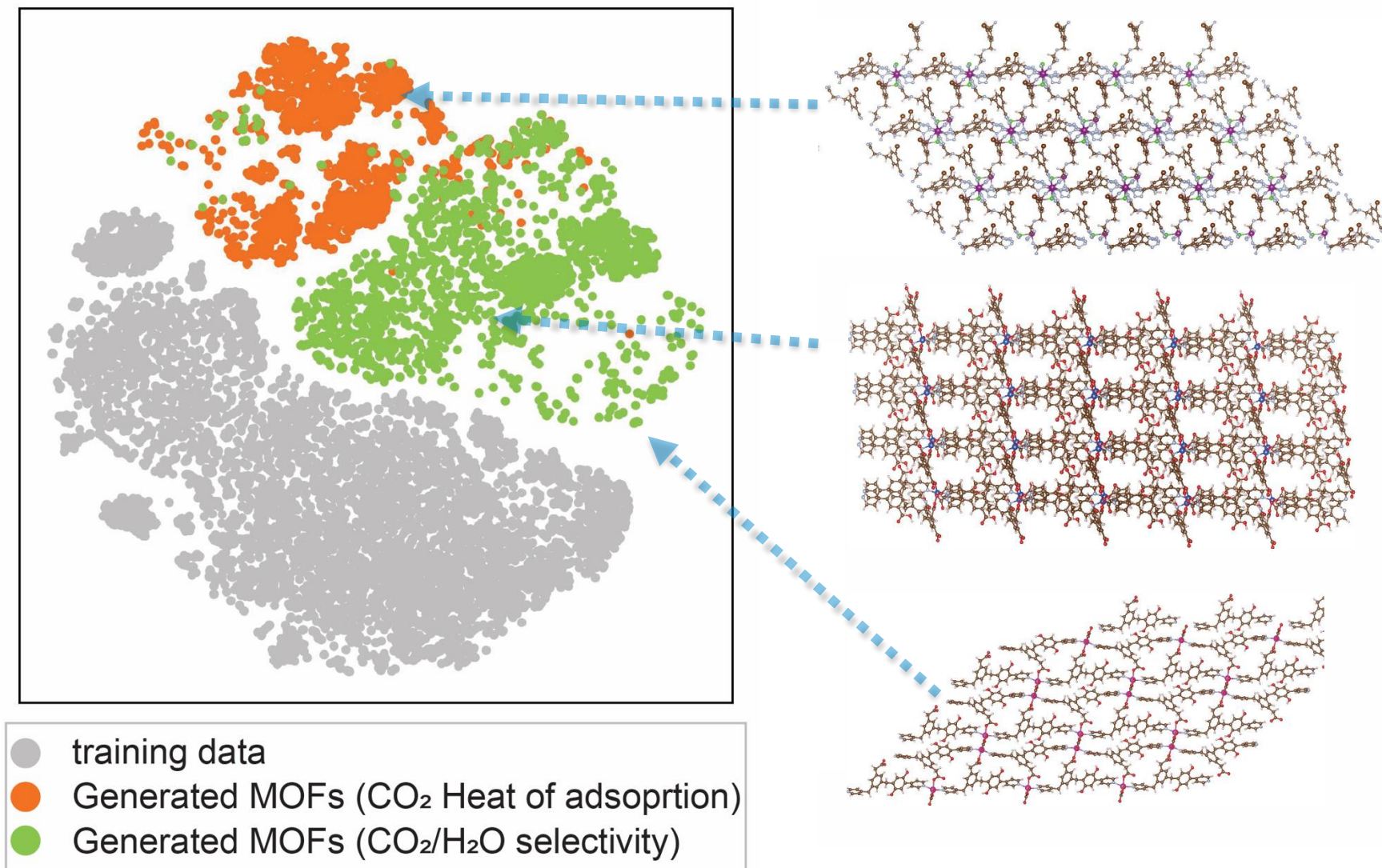
RL of Metal-Organic Frameworks



RL of Metal-Organic Frameworks



RL of Metal-Organic Frameworks



Experimental Optimisation Strategies

	Advantages	Disadvantages
High-throughput (Enumeration)	- Exhaustive search - Simple to implement and understand	- Inefficient for high-dimensional spaces - Maximises experiments and dataset size
Bayesian Optimisation	- Efficiently exploit data - Works with noisy and expensive evaluations	- Surrogate model & acquisition selection - Struggle with high-dimensional spaces
Reinforcement Learning	- Learns optimal policies through feedback - Can handle dynamic and complex environments	- Sample hungry - Slow to converge

Obstacles to Closed Loop Discovery

- **Materials complexity:** complex structures, compositions, processing sensitivity
- **Data quality and reliability:** errors and inconsistencies waste resources
- **Cost of automation:** major investment required in infrastructure and training
- **Adaptability:** systems and workflows may be difficult to reconfigure for new problems

Class Outcomes

1. Select an appropriate optimisation strategy for a given problem
2. Assess the impact of AI optimisation tools on materials research and discovery

Activity:

Closed-loop optimisation
